Econometrics Midterm

Multiple Choice · Econometrics 1–3 (Slides), Wooldridge Ch.1–6, Quiz 2 · 54 questions

1. An econometrician wants to learn the average hourly wage of all workers in Korea and collects data on 2,000 randomly selected workers. Which statement is correct?
   a. A sample is used only when the population is hypothetical
   b. The 2,000 workers are the population, because they are the units actually analyzed
   c. The 2,000 workers are the sample, and the estimates are statements about what the population of all Korean workers is like
   d. The estimates describe only the 2,000 workers and say nothing about other workers

2. Which statement best defines external validity?
   a. Whether the OLS estimator is unbiased within the sample used
   b. Whether the sample size is large enough to compute standard errors
   c. Whether the R-squared of the regression is high
   d. The validity of applying the conclusions of a study outside the context of that study

3. A researcher surveys only students at one private university in Seoul to estimate the average monthly smartphone spending of all Korean adults. What is the main concern?
   a. OLS cannot be computed because the population is too large
   b. The sample is unlikely to be representative of the population, so the conclusions may have weak external validity
   c. The estimates are automatically unbiased for all Korean adults because the students were chosen at random within the university
   d. The population becomes hypothetical, so no estimate is possible

4. Consider the model $\log(\text{price}) = \beta_0 + \beta_1 \text{subsidy} + \beta_2 \log(\text{sunlight}) + u$. How should β_1 be interpreted?
   a. The elasticity of price with respect to subsidy
   b. The slope from a simple regression of price on subsidy
   c. The change in log(price) from a one-unit increase in subsidy, holding log(sunlight) fixed (a partial slope)
   d. The total change in price from a one-unit increase in subsidy, including any change in sunlight it causes

5. In the model $\log(\text{price}) = \beta_0 + \beta_1 \text{subsidy} + \beta_2 \log(\text{sunlight}) + u$, which statement about subsidy is correct?
   a. Subsidy has a linear relationship with price, but not with log(price)
   b. Subsidy has a linear relationship with log(price), but not with price
   c. Subsidy has a linear relationship with both price and log(price)
   d. Subsidy has a linear relationship with neither price nor log(price)

6. In the model log(price) = $\beta_0 + \beta_1$ subsidy + β_2 log(sunlight) + u, which statement about sunlight is correct?

a. β_2 is the % change in price for a one-unit increase in sunlight

b. β_2 is the change in log(price) for a 1% increase in price

c. log(sunlight) has a linear relationship with log(price), so β_2 is approximately the % change in price for a 1% increase in sunlight, holding subsidy fixed

d. Sunlight has a linear relationship with price, so β_2 is the change in price for one more unit of sunlight

7. Suppose log(price) = $\beta_0 + \beta_1$ battery + β_2 battery^2 + u with $\beta_2 \neq 0$. Which statement is correct?

a. β_1 alone is the constant slope of battery for every value of battery

b. The model violates the assumption of linearity in parameters, so OLS cannot be used

c. Battery has a linear relationship with log(price) because the model is linear in parameters

d. Battery does not have a linear relationship with log(price); the effect of one more unit of battery depends on the current level of battery

8. Suppose log(price) = $\beta_0 + \beta_1$ battery + β_2 sunlight + β_3 battery× sunlight + u with $\beta_3 \neq 0$. Holding sunlight fixed, what is the effect on log(price) of one more unit of battery?

a. β_3· battery

b. β_1

c. $\beta_1 + \beta_3$

d. $\beta_1 + \beta_3$· sunlight

9. Which statement describes random sampling?

a. Every element has an equal probability of being assigned to the treatment or control group

b. The researcher chooses units that look typical of the population

c. Every element in the population has an equal probability of being selected into the sample

d. The sample contains more than half of the population

10. A researcher has a random sample of 500 workers and then flips a coin for each worker to decide who joins a job training program. The coin flip is an example of:

a. Random sampling

b. Perfect collinearity

c. Partialling out

d. Random assignment

11. Consider Model 2: wage = $\beta_0 + \beta_1$ educ + u, where ability is not included. People with better ability tend to get more education. Why can the OLS estimate of β_1 be biased even with a random sample?

a. There is no sample variation in educ

b. Ability goes into u and is correlated with educ, so the zero conditional mean condition E(u|educ) = 0 fails

c. Random sampling guarantees unbiasedness, so the estimate cannot be biased

d. The homoskedasticity assumption fails, which always causes bias

12. In Model 2, wage = $\beta_0 + \beta_1$ educ + u (ability omitted), suppose ability raises wage and is positively correlated with education. The OLS estimate of β_1 tends to:

a. Have a bias whose sign can never be determined
b. Understate the true return to education (downward bias)
c. Be unbiased, because ability is unobserved
d. Overstate the true return to education (upward bias)

13. The slides state: "OLS requires random sampling. It may also require random assignment." When estimating the effect of education on wages with Model 2 (wage = $\beta_0 + \beta_1$ educ + u), why would random assignment help?

a. Random assignment removes the need for any error term
b. Random assignment increases the R-squared of the regression
c. If educ were randomly assigned, it would be unrelated to ability and other factors in u, so the zero conditional mean condition would hold
d. Random assignment makes the sample representative of the population

14. A data set records the crime rate and the number of police officers for the same 150 cities in both 2020 and 2023. What type of data is this?

a. Pooled cross sections
b. Cross-sectional data
c. Panel (longitudinal) data
d. Time series data

15. A researcher draws a random sample of house prices in 1993 and a new, independent random sample of house prices in 1995 to evaluate a property tax change. What type of data is this?

a. Panel data
b. A single cross section
c. Time series data
d. Pooled cross sections

16. Which feature is typical of time series data but NOT of a cross section drawn by random sampling?

a. Observations are typically serially correlated, and their ordering conveys important information
b. Observations are more or less independent of each other
c. The ordering of observations does not matter
d. All observations are collected at the same point in time

17. In the simple regression y = $\beta_0 + \beta_1$ x + u, the OLS estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ are chosen to:

a. Minimize the sum of squared residuals
b. Make the fitted value equal to y for every observation
c. Minimize the sum of residuals
d. Minimize the sum of absolute residuals

18. In a simple regression of y on x, $R^2 = 0.25$. Which statement is correct?
    a. 25% of the observations lie exactly on the regression line
    b. 25% of the sample variation in y is explained by x
    c. x causes 25% of the change in y
    d. The estimated slope equals 0.25

19. The fitted regression is log(wage) = 0.584 + 0.083educ. How should 0.083 be interpreted?
    a. A 1% increase in education raises the wage by 8.3%
    b. One more year of education is associated with an approximately 8.3% higher wage
    c. One more year of education raises the wage by 0.083%
    d. One more year of education raises the wage by $0.083

20. Real earnings (in $1,000s) are regressed on train (= 1 if the person joined a randomly assigned job training program, 0 otherwise): earn = 4.55 + 1.79train. What does 1.79 measure?
    a. Nothing, because OLS cannot be used when x is binary
    b. Participants earn about 1.79% more than non-participants
    c. The average earnings of participants are $1,790
    d. The average treatment effect: participants earn about $1,790 more on average than non-participants

21. Under SLR.1–SLR.5, $\text{Var}(\hat{\beta}_1) = \sigma^2 / \text{SST}_x$. Which change makes $\hat{\beta}_1$ more precise (smaller variance)?
    a. More sample variation in x
    b. Fewer observations
    c. A larger error variance σ^2
    d. Replacing homoskedasticity with heteroskedasticity

22. In $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$, the OLS estimate $\hat{\beta}_1$ can also be obtained by:
    a. Regressing y on x_2 first, then adding x_1
    b. Regressing x_1 on x_2 (with a constant), then regressing y on the residuals from that regression
    c. Regressing x_2 on the residuals from a regression of y on x_1
    d. Averaging the slopes of the two simple regressions

23. Which regression violates MLR.3 (no perfect collinearity)?
    a. log(salary) on log(sales) and ceoten
    b. wage on educ and educ^2
    c. wage on educ and exper, where their correlation is 0.9
    d. wage on an intercept, male (= 1 if male) and female (= 1 if female)

24. The true model is $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$, but x_2 is omitted and y is regressed on x_1 only. In which case is the simple regression slope unbiased for β_1?
    a. When the sample size is very large
    b. When $\beta_2 = 0$ or when x_1 and x_2 are uncorrelated
    c. When the R-squared is high
    d. When x_2 is positively correlated with x_1 and $\beta_2 > 0$

25. Using 506 communities, the simple regression gives log(price) = 11.71 − 1.043 log(nox), while adding rooms gives log(price) = 9.23 − 0.718 log(nox) + 0.306 rooms. rooms raises price and is negatively correlated with log(nox). Which statement is correct?
    a. The simple regression is biased upward (toward zero), so −1.043 understates the true effect
    b. Adding rooms must always make the nox elasticity more negative
    c. The simple-regression elasticity −1.043 is biased downward (too negative), as predicted by $\beta_2 > 0$ and Corr(log(nox), rooms) < 0
    d. The two estimates differ only because of sampling error

26. High (but not perfect) correlation between x_1 and x_2 in a multiple regression mainly:
    a. Violates MLR.3, so OLS cannot be computed
    b. Increases the sampling variances of $\hat{\beta}_1$ and $\hat{\beta}_2$ but does not cause bias
    c. Causes $\hat{\beta}_1$ and $\hat{\beta}_2$ to be biased
    d. Inflates the variance of every coefficient, including an unrelated x_3

27. The Gauss–Markov theorem states that under MLR.1–MLR.5, the OLS estimators are:
    a. Exactly normally distributed in any sample
    b. The only unbiased estimators of the coefficients
    c. The best estimators even when the errors are heteroskedastic
    d. BLUE: they have the smallest variance among all linear unbiased estimators

28. In a regression, $\hat{\beta}_j = 0.80$ with $se(\hat{\beta}_j) = 0.10$. What is the t statistic for H_0: $\beta_j = 1$?
    a. 8.0
    b. −2.0
    c. 2.0
    d. −0.2

29. In a two-sided t test, the p-value is 0.07. Which statement is correct?
    a. H_0 is rejected at the 10% level but not at the 5% level
    b. H_0 is rejected at the 5% level
    c. H_0 is rejected at the 1% level
    d. The probability that H_0 is true is 7%

30. The 95% confidence interval for β_1 is [0.02, 0.10]. Which statement is correct?
    a. In repeated samples, intervals constructed this way contain the true β_1 95% of the time, and H_0: $\beta_1 = 0$ is rejected at the 5% level (two-sided)
    b. There is a 95% probability that $\hat{\beta}_1$ lies in [0.02, 0.10]
    c. β_1 equals 0.06 with 95% certainty
    d. H_0: $\beta_1 = 0$ cannot be rejected at the 5% level

31. To test H_0: $\beta_3 = 0$, $\beta_4 = 0$, $\beta_5 = 0$ jointly, which statement is correct?
    a. The F statistic can be negative when the restrictions help the fit
    b. Three separate t tests always give the same conclusion as the F test
    c. The restricted model always has a smaller SSR than the unrestricted model
    d. Use an F test comparing the SSR of the restricted and unrestricted models; F is large when the SSR rises a lot after imposing the restrictions

32. Which can cause the usual OLS t statistics to be invalid (not t-distributed under H_0)?
   a. A sample correlation of 0.95 between two explanatory variables in the model
   b. A low R-squared
   c. Heteroskedasticity, or omitting an important explanatory variable
   d. Having 500 observations instead of 5,000

33. For 32 chemical firms, rdintens = 0.472 + 0.321 log(sales) + 0.050 profmarg, where rdintens is R&D as a percentage of sales. If sales increase by 10%, rdintens changes by about:
   a. 3.21 percentage points
   b. 0.032 percentage points
   c. 32.1 percent
   d. 0.321 percentage points

34. Which statement about the consistency of OLS is correct?
   a. Consistency requires the errors to be normally distributed (MLR.6)
   b. Consistency requires homoskedasticity (MLR.5)
   c. A consistent estimator is unbiased in every finite sample
   d. Under MLR.1–MLR.4, the distribution of $\hat{\beta}_j$ collapses to the true β_j as $n \to \infty$, i.e. plim $\hat{\beta}_j = \beta_j$

35. pctstck = $\beta_0 + \beta_1$ funds + β_2 risktol + u satisfies MLR.1–MLR.4 (pctstck = % of pension in stocks, funds = number of mutual funds offered, risktol = risk tolerance). If funds and risktol are positively correlated and pctstck is regressed on funds only, the inconsistency in $\tilde{\beta}_1$ is:
   a. Zero, because the full model satisfies MLR.1–MLR.4
   b. Negative, because risktol is omitted
   c. It vanishes as the sample size grows
   d. Positive: plim $\tilde{\beta}_1 - \beta_1 = \beta_2\delta_1 > 0$, since $\beta_2 > 0$ and funds and risktol are positively correlated

36. cigs is the average number of cigarettes smoked per day. Can cigs have a normal distribution in the adult population, and what does this mean for OLS?
   a. Yes, by the central limit theorem every variable is normal
   b. No: cigs is nonnegative, piles up at zero (most adults do not smoke) and is discrete; still, with a large n, OLS t and F tests are approximately valid
   c. Yes, because cigs is an average
   d. No, so OLS estimates in a cigs equation are biased

37. Under MLR.1–MLR.5 but without normality (MLR.6), with a large sample:
   a. t tests become invalid no matter how large n is
   b. OLS becomes biased
   c. The usual t and F statistics are approximately valid, because $(\hat{\beta}_j - \beta_j)/\text{se}(\hat{\beta}_j)$ is approximately standard normal
   d. OLS standard errors cannot be computed

38. Under MLR.1–MLR.5, $\mathrm{se}(\hat{\beta}_j) \approx c_j / \sqrt{\{n\}}$. If the sample size is quadrupled, the standard error is approximately:

a. Quartered
b. Halved
c. Unchanged
d. Doubled

39. Crime model (n = 2,725): narr86 is regressed on pcnv, ptime86, qemp86 (restricted model). Its residuals are regressed on pcnv, avgsen, tottime, ptime86, qemp86, giving $R^2 = 0.0015$. To test H_0: avgsen and tottime have no effect:

a. $LM = nR^2 \approx 4.09$; compared with χ^2_2 (10% critical value 4.61), H_0 is not rejected at the 10% level
b. $LM = 0.0015$, so H_0 is rejected at the 1% level
c. $LM \approx 4.09$; compared with χ^2_5, H_0 is rejected
d. $LM = n(1 - R^2) \approx 2{,}721$, so H_0 is rejected

40. The dependent variable is changed from dollars to thousands of dollars (divided by 1,000). What happens?

a. Only the intercept changes
b. t statistics are divided by 1,000
c. R-squared falls
d. All coefficients and standard errors are divided by 1,000; t statistics and R-squared are unchanged

41. For 32 chemical firms, rdintens = 2.613 + 0.00030 sales − 0.0000000070 sales^2 (sales in $ millions). With salesbil = sales/1,000, the coefficients on salesbil and salesbil^2 become:

a. 0.00000030 and −0.0000000000070
b. 0.30 and −0.0000070
c. 0.30 and −0.0070, with the intercept and R-squared unchanged
d. 0.030 and −0.00070

42. For rdintens = 2.613 + 0.00030 sales − 0.0000000070 sales^2 (sales in $ millions), at what level of sales does the marginal effect of sales on rdintens become negative?

a. About 2,143
b. About 21,429, i.e. $21.4 billion
c. About 42,857
d. At sales = 0

43. A housing equation with dependent variable log(price) contains −0.545 rooms + 0.062 rooms^2 (other controls included). Moving from 6 to 7 rooms raises price by about:

a. 6.2%
b. 12.4%
c. −54.5%
d. 19.9%

44. In log(price) = 9.23 − 0.718 log(nox) + 0.306 rooms, what is the exact percentage change in price from one more room?

a. Exactly 30.6%

b. 0.306%

c. 3.06%

d. About 35.8%, = 100[exp(0.306) − 1]

45. In a regression where price and all explanatory variables are standardized (z-scores), the coefficient on nox is −0.340. Interpretation:

a. A 1 ppm increase in nox lowers price by 34%

b. A one standard deviation increase in nox lowers price by 0.34 standard deviation, holding the other variables fixed

c. A 1% increase in nox lowers price by 0.34%

d. nox explains 34% of the variation in price

46. log(wage) = 5.65 + 0.047educ + 0.00078educ· pareduc + 0.019exper + 0.010tenure (n = 722), where pareduc is total years of both parents' education. What is the estimated return to education when pareduc = 32?

a. 4.7%

b. 0.078%

c. About 7.2%

d. 2.5%

47. Which statement about adjusted R-squared is correct?

a. It always rises when a variable is added

b. It is always larger than R-squared

c. It can compare a model for y with a model for log(y)

d. It can fall when a variable is added; it rises only if the new variable's |t| exceeds 1

48. College GPA equation (n = 4,137): predicted colgpa = 2.70 at sat = 1200, hsperc = 30, hsize = 5, with se($\hat{y}^0$) = 0.020 and $\hat{\sigma}$ = 0.560. Which is correct?

a. The prediction interval for one student is narrower than the confidence interval for the mean

b. Both 95% intervals are 2.66 to 2.74

c. The prediction interval shrinks to zero width as n grows

d. The 95% CI for expected colgpa is about 2.66–2.74, but the 95% prediction interval for one student is about 1.60–3.80, because it also includes the error variance σ^2

49. The dependent variable is log(y). Using exp(predicted log(y)) as the prediction of y:

a. Systematically underestimates E(y|x); it should be scaled up by $\hat{\alpha}_0$, e.g. exp($\hat{\sigma}^2$/2) under normality or the average of exp($\hat{u}_i$)

b. Gives an unbiased prediction of y

c. Overestimates E(y|x)

d. Is valid only when R-squared is high

50. Palm oil production is estimated as: production = 243.63 − 3.34 tax − 0.61 fuel − 0.03 fuel^2. Production is the amount of palm oil produced, tax is the tax rate (%), and fuel is the fuel price ($). What is the predicted amount of palm oil produced when the tax rate is 5% and the fuel price is $7?

a. 222.45
b. 221.19
c. 226.93
d. 222.66

51. The palm oil equation above is estimated with an intercept and three regressors (tax, fuel, fuel^2). What are the degrees of freedom when the sample size is 85?

a. 84
b. 82
c. 81
d. 80

52. Worker productivity is estimated as: productivity = 3.36 + 0.06 log(training) + 2.53 log(experience) + 0.03 log(training) × log(experience). Productivity is shoes made per hour, training is training hours, experience is years of experience (log = natural log). What is the predicted number of shoes the worker makes in one hour with 2000 hours of training and 17 years of experience?

a. 6.79
b. 10.98
c. 11.63
d. 11.30

53. Vaccination is estimated as: log(vaccination) = 0.34 + 0.15 reminder + 0.23 log(ambassador). Reminder is the number of reminders sent and ambassador is the number of ambassadors who are information hubs (log = natural log). What is the predicted amount of log(vaccination) when 1 reminder is sent and there are five ambassadors?

a. 0.65
b. 0.86
c. 1.64
d. 0.49

54. Vaccination is estimated as: log(vaccination) = 0.34 + 0.15 reminder + 0.23 log(ambassador). Reminder is the number of reminders sent and ambassador is the number of ambassadors who are information hubs (log = natural log). What is the predicted amount of vaccination when 1 reminder is sent and there are five ambassadors?

a. 4.47
b. 0.86
c. 2.36
d. 5.16

Econometrics Midterm — Answers

1. c
A sample is the subset of the population chosen for analysis, and we study it to learn what the population is like. An estimate is a judgment about the population based on the sample.

2. d
External validity asks whether a study's conclusions can validly be applied outside the context of that study. It is directly tied to whether the sample represents the population.

3. b
Students at one university are unlikely to represent all Korean adults. We must ask whether the sample represents the population, and if it does not, the conclusions are hard to apply to that population (weak external validity).

4. c
In multiple regression β_1 is not a simple slope but a partial slope: the effect with the other regressor (log(sunlight)) held fixed. The dependent variable is log(price), so the units are also in terms of log(price).

5. b
log(price) is a linear function of subsidy, so that relationship is linear. But price = $\exp(\beta_0 + \beta_1$ subsidy + ...), so price and subsidy are not linearly related.

6. c
Log-log form: log(sunlight) and log(price) are linearly related, and β_2 is an elasticity (a 1% increase leads to about a β_2% change). Sunlight and price themselves are not linearly related.

7. d
Δ log(price)/Δ battery $\approx \beta_1 + 2\beta_2$ battery depends on the level of battery, so the relationship is not linear. The model is still linear in the parameters, so OLS can estimate it.

8. d
Because battery and sunlight interact, the effect of battery is $\beta_1 + \beta_3 \cdot$ sunlight and depends on the level of sunlight. So β_1 alone is not the slope of battery, and battery and log(price) are not linearly related.

9. c
Random sampling means every element of the population has the same probability of being drawn into the sample. Equal probability of group assignment is random assignment.

10. d
Drawing the sample is random sampling. Splitting the drawn people into treatment and control groups by a coin flip is random assignment.

11. b
If ability is left out of the model it goes into the error u. If ability and education are correlated, E(u|educ) = 0 (zero conditional mean) fails, so β_1 is biased even under random sampling.

12. d
Omitted variable bias = (effect of ability β_2) × (direction of the correlation between educ and abil). Both are positive, so the bias is positive, that is, the estimate is overstated.

13. c
With random assignment, education is not systematically related to factors in the error such as ability, so E(u|educ) = 0 holds and β_1 is unbiased. (Representativeness is the job of random sampling.)

14. c
Observing the same units (cities) repeatedly over time gives panel data. It lets us control for unobserved characteristics that do not change over time.

15. d
Combining two cross sections drawn independently at two points in time gives a pooled cross section. The same households are not followed, so it is not a panel. It is often used to compare before and after a policy change.

16. a
In a time series the observations are serially correlated and their order matters (trend, seasonality). In a randomly sampled cross section the observations are roughly independent and their order carries no information.

17. a
OLS (ordinary least squares) minimizes the sum of squared residuals $\Sigma\hat{u}_i^2$. As a result the residuals always sum to zero (an algebraic property).

18. b
R^2 = SSE/SST = 1 − SSR/SST is the proportion of the sample variation in y explained by x. It does not mean causality or the size of the slope.

19. b
Log-level (semi-log) form: a one-unit increase in x changes y by about $100\beta_1$ %. 0.083 × 100 = 8.3%.

20. d
If x is 0/1, the slope is the difference between the two group means. Under random assignment OLS is an unbiased estimator of the treatment effect. The non-participant mean is $4,550, so 1,790 is about a 39.3% increase.

21. a
The variance rises with the error variance σ^2 and falls with the sample variation in x, SST_x.

22. b
Partialling out: regress y on the residual of x_1 after removing the part explained by x_2 (the part uncorrelated with x_2), and the slope is β_1.

23. d
With an intercept, male + female = 1 is an exact linear relationship, so there is perfect collinearity. $educ^2$ is not a linear function of educ, and a high correlation (0.9) is multicollinearity, not a violation of MLR.3.

24. b
Omitted variable bias = $\beta_2 \times \delta_1$. If the omitted variable is irrelevant ($\beta_2 = 0$) or uncorrelated with x_1 ($\delta_1 = 0$), there is no bias.

25. c
Sign of the omitted variable bias = (effect of rooms β_2, +) × (correlation between log(nox) and rooms, −) = negative, so the bias is downward. Indeed −1.043 is more negative than −0.718. (Wooldridge problem, HPRICE2)

26. b
Multicollinearity only inflates the variances of the related variables (diagnosed with VIF) and unbiasedness is preserved. It is not a violation of MLR.3 in the strict sense.

27. d
Under MLR.1–5, OLS has the smallest variance among linear unbiased estimators (BLUE). With heteroskedasticity a better estimator exists. Normality requires MLR.6.

28. b
$t = (\hat{\beta}_j - a_j)/se(\hat{\beta}_j) = (0.80 - 1)/0.10 = -2.0$. Subtract the hypothesized value and divide by the standard error.

29. a
The p-value is the smallest significance level at which H_0 can be rejected. Reject only when $p < \alpha$: $0.07 < 0.10$ but $0.07 > 0.05$.

30. a
The confidence interval's endpoints are random (the parameter is not). Zero lies outside the 95% interval, so the two-sided 5% test rejects $\beta_1 = 0$.

31. d
$F = [(SSR_r - SSR_{ur})/q] \,/\, [SSR_{ur}/(n-k-1)]$. Imposing restrictions can only raise SSR, so $F \geq 0$, and the individual t tests and the joint test can give different conclusions.

32. c
Heteroskedasticity violates MLR.5 and omitting an important variable violates MLR.4 (zero conditional mean), so the t distribution does not hold. A high correlation (0.95) between regressors is not a CLM violation (only perfect collinearity is ruled out). (Wooldridge Problem 4.1)

33. b
Level-log form: $\Delta y \approx (\beta_1/100)(\%\Delta x) = (0.321/100) \times 10 \approx 0.032$. rdintens is measured in percent, so this is a percentage-point change. (Wooldridge problem, RDCHEM)

34. d
Consistency: as the sample grows the distribution of the estimator concentrates on the true value. MLR.1–MLR.4 suffice, and even the weaker MLR.4′ ($E(u) = 0$, $Cov(x_j, u) = 0$) is enough. Normality and homoskedasticity are not needed.

35. d
Inconsistency = $\beta_2 \times \delta_1$ with $\delta_1 = Cov(funds, risktol)/Var(funds)$. A higher risk tolerance means a larger stock share, so $\beta_2 > 0$, and the correlation is positive, giving a positive inconsistency. It does not vanish as the sample grows. (Wooldridge Problem 5.2)

36. b
The normal distribution is continuous and symmetric and can take negative values. cigs has a mass at zero, so it is not normal. But normality (MLR.6) is unrelated to unbiasedness, and in large samples inference is approximately valid thanks to asymptotic normality. (Wooldridge Problem 5.3)

37. c
Asymptotic normality (central limit theorem): even without normality, t and F tests and confidence intervals are approximately valid in large samples. Homoskedasticity (MLR.5) is still needed, though. With heteroskedasticity the usual standard errors are wrong even in large samples.

38. b
Standard errors shrink at the rate $1/\sqrt{n}$. Since $\sqrt{4} = 2$, they fall to about half.

39. a
LM = $n \cdot R^2_u = 2{,}725 \times 0.0015 \approx 4.09$, the degrees of freedom are the number of restrictions q = 2, and the p-value ≈ 0.13. Regress the restricted-model residuals on all of the regressors. (Wooldridge Example 5.3)

40. d
Changing the units of y changes the coefficients and standard errors by the same factor. So the t statistics, p-values and R^2 do not change (same statistical conclusions).

41. c
salesbil coefficient = 0.00030 × 1,000 = 0.30, and salesbil^2 = sales^2/1,000,000, so its coefficient = −0.0000000070 × 10^6 = −0.0070. This gives a form that is easy to report. (Wooldridge Problem 6.3)

42. b
Turning point x^* = $\hat{\beta}_1/(2|\hat{\beta}_2|)$ = 0.00030/(2 × 0.0000000070) ≈ 21,428.57 ($ millions). Beyond it the marginal effect becomes negative. (Wooldridge Problem 6.3)

43. d
%Δ price ≈ 100[−0.545 + 2(0.062)(6)] ≈ 19.9%. The turning point is 0.545/(2 × 0.062) ≈ 4.4 rooms, and only about 1% of the communities in the sample have fewer rooms than that. (Wooldridge Example 6.2, HPRICE2)

44. d
The 100β approximation for log-level is accurate only for small changes. The exact value is 100[exp(βΔ x) − 1] = 100(1.358 − 1) ≈ 35.8%.

45. b
A standardized (beta) coefficient shows how many standard deviations y changes when x changes by one standard deviation, so it lets us compare the effects of variables measured in different units. (Wooldridge Example 6.1)

46. c
Δ log(wage)/Δ educ = 0.047 + 0.00078 × pareduc = 0.047 + 0.025 ≈ 0.072. The higher the parents' education, the larger the return to education. (Wooldridge Problem 6.4, WAGE2)

47. d
Adjusted R^2 = 1 − [SSR/(n−k−1)]/[SST/(n−1)] penalizes through the degrees of freedom. R^2 never falls when variables are added, but adjusted R^2 can fall. It cannot be compared across different dependent variables (y vs log y).

48. d
Confidence interval for the mean: 2.70 ± 1.96(0.020). Prediction interval for one individual: se($\hat{e}^0$) = $\sqrt{\{0.020^2 + 0.560^2\}}$ ≈ 0.560, giving 2.70 ± 1.96(0.560). The error variance does not shrink as n grows. (Wooldridge Examples 6.5–6.6, GPA2)

49. a
E(y|x) = exp(σ^2/2)·exp(xβ), so using only exp(log ŷ) underpredicts. Multiply by $\hat{\alpha}_0 > 1$ to correct it. (Wooldridge Section 6.4, Example 6.7)

50. b
production = 243.63 − 3.34(5) − 0.61(7) − 0.03$(7)^2$ = 243.63 − 16.70 − 4.27 − 1.47 = 221.19.

51. c
df = n − k − 1 = 85 − 3 − 1 = 81.

52. c
ln(2000) = 7.6009 and ln(17) = 2.8332
productivity = 3.36 + 0.06(7.6009) + 2.53(2.8332) + 0.03(7.6009)(2.8332) = 3.36 + 0.4561 + 7.1680 + 0.6460 = 11.63
log base 10 would give 6.79, so log means ln.

53. b
log(vaccination) = 0.34 + 0.15(1) + 0.23 ln(5) = 0.34 + 0.15 + 0.23(1.6094) = 0.86.

54. c
vaccination = exp(0.8602) = $e^{0.8602}$ = 2.36.

Econometrics Midterm

Short Answer · Econometrics 1–3 (Slides), Wooldridge Ch.1–6, Quiz 2 · 37 questions

1. What is the difference between a population and a sample?

2. Why do econometricians usually study a sample instead of the population, and what is the goal of studying the sample?

3. Define external validity and state which property of the sample matters for it.

4. In $\log(\text{price}) = \beta_0 + \beta_1 \text{subsidy} + \beta_2 \log(\text{sunlight}) + u$, why is β_1 called a partial slope rather than simply "the slope"?

5. In $\log(\text{price}) = \beta_0 + \beta_1 \text{battery} + \beta_2 \text{sunlight} + \beta_3 \text{battery} \times \text{sunlight} + u$ with $\beta_3 \neq 0$, why does battery not have a linear relationship with log(price)?

6. What is the difference between random sampling and random assignment?

7. For Model 2, $\text{wage} = \beta_0 + \beta_1 \text{educ} + u$ (ability not included), explain why the zero conditional mean condition may be violated.

8. What is a control variable, and why is ability added in Model 1, wage = $\beta_0 + \beta_1$ educ + β_2 abil + u?

9. What does "ceteris paribus" mean, and why is it important in econometrics?

10. What is panel (longitudinal) data, and what is one advantage of it?

11. What does OLS minimize, and what is a residual?

12. What does R-squared measure in a regression?

13. What does the homoskedasticity assumption (SLR.5) state?

14. What does it mean that the OLS estimator β_1 is unbiased?

15. When does omitting a relevant-looking variable x_2 NOT cause bias in the estimated coefficient on x_1?

16. What is the main consequence of multicollinearity (high but not perfect correlation among regressors) for OLS?

17. State the Gauss–Markov theorem.

18. Define the p-value of a hypothesis test.

19. Does a statistically significant coefficient always matter in practice? Answer in 1–2 sentences.

20. Interpret a 95% confidence interval for β_j.

21. Which can make the usual OLS t statistics invalid (no t distribution under H_0): (i) heteroskedasticity, (ii) a sample correlation of 0.95 between two regressors in the model, (iii) omitting an important explanatory variable? Explain briefly.

22. What does it mean that an estimator is consistent, and which assumptions are enough for OLS to be consistent?

23. pctstck = $\beta_0 + \beta_1$ funds + β_2 risktol + u satisfies MLR.1–MLR.4 (risktol = risk tolerance). If funds and risktol are positively correlated and pctstck is regressed on funds only, what is the sign of the inconsistency in $\tilde{\beta_1}$, and why?

24. Can cigs (average cigarettes smoked per day) have a normal distribution in the adult population? What does this imply for OLS inference?

25. How is the LM (n-R-squared) statistic for testing q exclusion restrictions computed, and what distribution does it follow?

26. If the dependent variable is divided by 1,000 (dollars → thousands of dollars), what happens to the OLS coefficients, standard errors, t statistics and R-squared?

27. What is a beta (standardized) coefficient, and why is it useful?

28. For rdintens = 2.613 + 0.00030 sales − 0.0000000070 sales^2 (sales in $ millions), give the turning-point formula for a quadratic and find where the marginal effect of sales becomes negative.

29. In log(wage) = $\beta_0 + \beta_1$ educ + β_2 educ· pareduc + β_3 exper + β_4 tenure + u (pareduc = both parents' education), what is the return to one more year of education, and what does β_2 measure?

30. Why is adjusted R-squared used, and when does adding a variable increase it?

31. Why is a 95% prediction interval for one individual outcome y^0 much wider than the 95% confidence interval for $E(y|x = x^0)$?

32. When the dependent variable is log(y), why is exp(predicted log(y)) not a good prediction of y, and how is it fixed?

33. Palm oil production is estimated as: production = 243.63 − 3.34 tax − 0.61 fuel − 0.03 fuel^2. Production is the amount of palm oil produced, tax is the tax rate (%), and fuel is the fuel price ($). What is the predicted amount of palm oil produced when the tax rate is 5% and the fuel price is $7? Write the equation with the numbers substituted (two decimal places).

34. The palm oil equation above is estimated with an intercept and three regressors (tax, fuel, fuel^2). What are the degrees of freedom when the sample size is 85? Write the formula and the numbers.

35. Worker productivity is estimated as: productivity = 3.36 + 0.06 log(training) + 2.53 log(experience) + 0.03 log(training) × log(experience). Productivity is shoes made per hour, training is training hours, experience is years of experience (log = natural log). What is the predicted number of shoes the worker makes in one hour with 2000 hours of training and 17 years of experience? Write the equation with the numbers substituted (two decimal places).

36. Vaccination is estimated as: log(vaccination) = 0.34 + 0.15 reminder + 0.23 log(ambassador). Reminder is the number of reminders sent and ambassador is the number of ambassadors who are information hubs (log = natural log). What is the predicted amount of log(vaccination) when 1 reminder is sent and there are five ambassadors? Write the equation with the numbers substituted (two decimal places).

37. Vaccination is estimated as: log(vaccination) = 0.34 + 0.15 reminder + 0.23 log(ambassador). Reminder is the number of reminders sent and ambassador is the number of ambassadors who are information hubs (log = natural log). What is the predicted amount of vaccination when 1 reminder is sent and there are five ambassadors? Write the equation with the numbers substituted (two decimal places).

Econometrics Midterm — Model answers

1. Econ 1
A population is the entire set of individuals, objects or data points we want to study, while a sample is a subset of the population that is selected for analysis.
Key results: entire · subset · selected

2. Econ 1
Studying the whole population is often too costly or impossible, so we study a sample
the goal is to find out what the population is like from the sample.
Key results: costly · impossible · population is like

3. Econ 1
External validity is the validity of applying the conclusions of a study outside the context of that study
it depends on whether the sample is representative of the population.
Key results: outside · conclusions · representative

4. Econ 2
β_1 is the change in log(price) from a one-unit increase in subsidy while holding the other variables (log(sunlight)) fixed, so it isolates the effect of subsidy alone.
Key results: holding · other variables · one-unit

5. Econ 2
Because battery and sunlight interact, the effect of battery on log(price) is $\beta_1 + \beta_3 \cdot$sunlight, which depends on the level of sunlight
the slope is not constant.
Key results: interact · depends · not constant

6. Econ 3
In random sampling every element of the population has an equal probability of being selected
in random assignment every element has an equal probability of being assigned to a group.
Key results: equal probability · selected · assigned

7. Econ 3
Ability is omitted, so it goes into the error u
because people with better ability get more education, educ is correlated with u, so $E(u|educ) = 0$ fails and β_1 is estimated with bias.
Key results: ability · error · correlated

8. Econ 3
A control variable is included to hold another factor fixed so it no longer sits in the error
adding ability helps satisfy the zero conditional mean condition and removes the bias in β_1.
Key results: hold · zero conditional mean · bias

9. Ch.1
Ceteris paribus means "other relevant factors being equal"
most economic questions ask for the causal effect of one variable holding all other factors fixed.
Key results: other factors · equal · causal

10. Ch.1
Panel data follow the same cross-sectional units over time
they can account for time-invariant unobservables and model lagged responses.
Key results: same · over time · time-invariant

11. Ch.2
OLS chooses the estimates that minimize the sum of squared residuals
a residual is the actual value y_i minus the fitted value $\hat{y}_i$.
Key results: sum of squared residuals · actual · fitted

12. Ch.2
R^2 = SSE/SST is the fraction of the sample variation in y that is explained by the explanatory variable(s).
Key results: fraction · variation · explained

13. Ch.2
Homoskedasticity means the error u has the same (constant) variance for any value of x: $Var(u|x) = \sigma^2$.
Key results: variance · error · same

14. Ch.2
Unbiasedness means $E(\hat{\beta}_1) = \beta_1$: if sampling and estimation were repeated many times, the estimates would on average equal the true population value, though one estimate may differ.
Key results: on average · repeated · true

15. Ch.3
There is no omitted variable bias if the omitted variable is irrelevant (its coefficient is zero) or uncorrelated with x_1.
Key results: irrelevant · uncorrelated

16. Ch.3
Multicollinearity increases the sampling variance of the affected slope estimators, but OLS remains unbiased
it can be detected with variance inflation factors.
Key results: variance · increase · unbiased

17. Ch.3
Under MLR.1–MLR.5, OLS is BLUE (best linear unbiased estimator): it has the smallest variance among all linear unbiased estimators.
Key results: BLUE · smallest variance · MLR.1

18. Ch.4
The p-value is the smallest significance level at which the null hypothesis would be rejected
we reject H_0 when the p-value is below the chosen level.
Key results: smallest · significance level · reject

19. Ch.4
Not necessarily: statistical significance only shows the effect differs from zero, while economic (practical) significance depends on the magnitude of the coefficient.
Key results: not necessarily · magnitude · economic

20. Ch.4
In repeated samples, intervals built this way contain the true population coefficient in 95% of cases
the bounds are random, not the parameter.
Key results: repeated · contain · true

21. Ch.4
Heteroskedasticity (i) and an omitted important variable (iii) violate the CLM assumptions, so the t statistics are invalid
a high correlation of 0.95 (ii) is not a CLM violation (only perfect collinearity is ruled out).
Key results: heteroskedasticity · omitted · correlation

22. Ch.5
An estimator is consistent if, as the sample size grows, its distribution collapses to the true parameter (plim $\hat{\beta}_j = \beta_j$)
OLS is consistent under MLR.1–MLR.4 (even zero correlation between each x_j and u is enough).
Key results: sample size · collapse · true · MLR.4

23. Ch.5
The inconsistency is positive: plim $\tilde{\beta}_1 - \beta_1 = \beta_2\delta_1$, with $\beta_2 > 0$ (higher risktol raises pctstck) and $\delta_1 > 0$ because funds and risktol are positively correlated
it does not disappear as n grows.
Key results: positive · risktol · correlated · not disappear

24. Ch.5
No: most adults smoke zero cigarettes and cigs is nonnegative and discrete, so it is not normal
with a large sample, asymptotic normality still makes the t and F tests approximately valid.
Key results: zero · nonnegative · not normal · large sample

25. Ch.5
Estimate the restricted model, regress its residuals on all the explanatory variables, and compute LM = nR^2 from that auxiliary regression
under H_0 it is approximately chi-square with q degrees of freedom.
Key results: restricted · residuals · nR · chi-square

26. Ch.6
All coefficients and standard errors are divided by 1,000, while the t statistics and R-squared stay unchanged
only the units change, not the statistical conclusions.
Key results: divided · unchanged · t statistic · R-squared

27. Ch.6
A beta coefficient comes from regressing the standardized y on standardized x's
it is the change in y, in standard deviations, from a one standard deviation increase in x_j, so effects measured in different units can be compared.
Key results: standardized · standard deviation · units · compare

28. Ch.6
Turning point $\hat{x}^* = \hat{\beta}_1/(2|\hat{\beta}_2|) = 0.00030/(2 \times 0.0000000070) \approx 21{,}428.57$, i.e. sales of about $21.4 billion beyond that the estimated marginal effect of sales on rdintens is negative.

29. Ch.6
The return is $\beta_1 + \beta_2 \cdot$pareduc, so it depends on parents' education
the interaction coefficient β_2 is how much the return changes for one more year of pareduc (estimate 0.00078: at pareduc = 32 the return is about 7.2%).
Key results: depends · parents · interaction

30. Ch.6
R-squared never falls when a variable is added, so adjusted R-squared imposes a penalty for extra regressors using degrees of freedom
adding a variable raises it only if that variable's t statistic exceeds one in absolute value.
Key results: never falls · penalty · degrees of freedom · t statistic

31. Ch.6
The confidence interval reflects only the sampling variance of $\hat{y}^0$, while the prediction interval also includes the variance of the unobserved error for an individual outcome (σ^2), which does not shrink as n grows (e.g. colgpa: CI 2.66–2.74 vs PI 1.60–3.80).
Key results: error · variance · individual

32. Ch.6
$\exp(\log \hat{y})$ systematically underestimates $E(y|x)$
we multiply it by $\hat{\alpha}_0 = \exp(\sigma^2/2)$ under normal errors, or by the average of $\exp(\hat{u}_i)$ from the residuals.
Key results: underestimate · multiply · residuals

33. Ch.6
production $= 243.63 - 3.34(5) - 0.61(7) - 0.03(7)^2$
$= 243.63 - 16.70 - 4.27 - 1.47 = 221.19$
Key results: 221.19 · 16.7 · 4.27 · 1.47

34. Ch.3
df $= n - k - 1 = 85 - 3 - 1 = 81$
Key results: 81 · n − k − 1 · 85 − 3 − 1

35. Ch.6
ln(2000) $= 7.6009$ and $\ln(17) = 2.8332$
productivity $= 3.36 + 0.06(7.6009) + 2.53(2.8332) + 0.03(7.6009)(2.8332)$
$= 3.36 + 0.4561 + 7.1680 + 0.6460 = 11.63$
log base 10 would give 6.79, so log means ln
Key results: 11.63 · 7.60 · 2.83 · 7.17 · 0.65

36. Ch.6
log(vaccination) $= 0.34 + 0.15(1) + 0.23 \ln(5) = 0.34 + 0.15 + 0.23(1.6094) = 0.86$
Key results: 0.86 · 0.37 · 1.61

37. Ch.6
vaccination $= \exp(0.8602) = e^{0.8602} = 2.36$
Key results: 2.36 · 0.86

Econometrics Midterm

R Functions · W1, W2, W5, W6 · 31 questions

1. Read the code. What does table(x) return, and what does sort(..., decreasing = TRUE) do with it?

```
x <- c(2, 3, 3, 5, 3, 2)
table(x)
sort(table(x), decreasing = TRUE)
```

2. Read the code from Lab2.R. What does read.csv() do, and what do the arguments header = TRUE and na.strings = "." do?

```
practice <- read.csv("Lab2_Practice.csv", header=TRUE, na.strings=".")
```

3. Read the code from Lab2.R. What do attach(practice) and detach(practice) do?

```
attach(practice)
mean(COMPETENCE)
detach(practice)
```

4. For breaks, the first line returns 0, 1, 2, 4.5 and 12 and the second returns 3.5. What does each function compute?

```
quantile(breaks)
IQR(breaks)
```

5. Read the code from Lab2.R. What does boxplot(grade) draw, and what does abline(h = median(grade), col = "red") add to it?

```
boxplot(grade)
abline(h=median(grade), col="red")
```

6. Read the code from Lab2.R. What does vioplot() draw, and why must install.packages("vioplot") and library("vioplot") be run first?

```
install.packages("vioplot")
library("vioplot")
vioplot(competence, breaks, grade)
```

7. Read the code from Lab2.R. What does par(mfrow = c(1, 2)) do to the plots that follow it?

```
par(mfrow = c(1, 2))
boxplot(grade)
vioplot(grade)
```

8. What does source() do, and what does the argument echo = TRUE add?

```
source("~/R_w2.R", echo = TRUE)
```

9. Read the code from Lab2.R (competence has 27 values). What does the first line compute, step by step, and which built-in function gives the same value?

```
sum((competence-mean(competence))^2)/(length(competence)-1)
var(competence)
```

10. In mtcars, am = 0 marks cars with an automatic transmission. What does mpg[am == 0] select, and what does the whole line compute?

```
attach(mtcars)
mean(mpg[am==0])
```

11. Read the function from Lab4.R. What does to12() do with its input, and what do as.integer() and paste() contribute to it?

```
to12 <- function(str){
  hour=as.integer(str)
  if (hour>12) {
    return(paste(hour-12, "PM"))
  } else {
    return(paste(hour, "AM"))
  }
}
to12("22")
```

12. In Lab4.R, hour = "13" is stored as text. Why does paste(hour - 12, "PM") stop with an error, and how does to12() avoid the problem?

```
hour="13"
if(as.integer(hour)>12) {
  print(paste(hour-12, "PM"))
}
```

13. Read the code from Lab4.R. What does str_detect(str, fixed(",")) return, what does fixed() do, and what do the two calls return?

```
library(stringr)
hasMultipleLoc <- function(str){
  str_detect(str, fixed(","))
}
hasMultipleLoc("Seoul, Suwon")
hasMultipleLoc("Seoul")
```

14. Read the code from Lab4.R. What does sapply() do here, and why is it used instead of calling to12(data$openingHour) directly?

```
(openingHour <- sapply(data$openingHour, to12))
```

15. Read the code from Lab4.R. What does ifelse() do in this line, and what ends up in the new column?

```
(locationText <- ifelse(sapply(data$location, hasMultipleLoc), "Yes", "No"))
data$hasMultipleLoc = locationText
```

16. Read the code from Lab4.R. What does each of the three functions do?

```
library(readxl)
data = read_excel("Café.xlsx")
data <- as.data.frame(data)
colnames(data) = c("type", "location", "openingHour", "closingHour")
```

17. Read the code from regression.R. What does lm() return here, and what does abline() draw when it is given that result?

```
plot(perform ~ eye)
eye_regr = lm(perform ~ eye)
abline(eye_regr, col = "blue")
```

18. Read the code from regression.R. What does summary() report for an lm model, and what does $r.squared extract?

```
eye_summary = summary(eye_regr)
eye_summary
eye_summary$r.squared
```

19. Read the code from regression.R. What does confint() return, and what changes when level = 0.99 is added?

```
confint(eye_regr)
confint(eye_regr, level = 0.99)
```

20. Read the code from regression.R. What does cor() compute, why is it squared, and what is round(..., 16) for?

```
(skill_coef = cor(perform, skill)^2)
round(skill_summary$r.squared - skill_coef, 16)
```

21. Read the code from regression.R. What does predict() do with the data frame it is given, and what do the arguments interval and level change?

```
predict(mult_lm, data.frame(skill=c(0.5,1.5), ravens=c(1.5,0.5)),
interval="confidence", level=0.95)
```

22. Read the code from regression.R. What does each of seq(), polygon() and lines() do in drawing the confidence band?

```
points <- seq(min(eye)-3, max(eye)+3, length.out=10000)
pred <- predict(eye_regr, data.frame(eye=points), interval="confidence")
polygon(c(rev(points), points), c(rev(pred[ ,3]), pred[ ,2]), col=rgb(0,0,0,0.1),
border=NA)
lines(points, pred[ ,3], lty='dashed', col='green')
```

23. Read the code from the simulation script. What do set.seed() and rnorm() do?

```
set.seed(4201)
fertilizer = rnorm(sampleSize, 10, 2)
yield = intercept + slope*fertilizer + rnorm(sampleSize, 0, 1)
```

24. Read the simulation loop (sampleCnt = 10,000). What does the for loop do, and what do coef[2,1], coef[2,3] and coef[2,4] contain?

```
for(rowIdx in 1:sampleCnt) {
  fertilizer=rnorm(sampleSize,10,2)
  yield=intercept+slope*fertilizer+rnorm(sampleSize,0,1)
  coef=summary(lm(yield~fertilizer))$coefficients
  estimator[rowIdx]=coef[2,1]
  t[rowIdx]=coef[2,3]
  pValue[rowIdx]=coef[2,4]
}
```

25. Read the code from the simulation script. What does each line do, and what does the share of "Reject" estimate here (the true slope is 0.2)?

```
rejection = pValue <= alpha
rejection = ifelse(rejection, "Reject", "Fail to Reject")
(result = table(rejection))
barplot(result, main = "Null Hypothesis Rejection")
```

26. Read the confidence-interval loop. What do data.frame(), results[rowIdx, "slope"], interval["fertilizer", "2.5 %"] and && do?

```
results = data.frame(index=1:sampleCnt)
results[rowIdx,"slope"] = fer_regr$coefficients[2]
interval = confint(fer_regr)
lowerBound = interval["fertilizer","2.5 %"]
upperBound = interval["fertilizer","97.5 %"]
inside = lowerBound<=slope && slope<=upperBound
```

27. Read the plotting code from the simulation script. What does each layer add to the chart of the 20 confidence intervals?

```
ggplot(results, aes(x=slope, y=index)) + geom_point(size=3.5) +
  geom_errorbarh(aes(xmin = lower_bound, xmax = upper_bound)) +
  geom_vline(xintercept = 2, color = "green")
```

28. What does as.factor(session) do in this regression, and what does lm() then create from it?

```
model <- lm(y ~ x + as.factor(session), data = df)
```

29. What terms does each of the three formulas put into the model?

```
lm(y ~ x1:x2)
lm(y ~ x1*x2)
lm(y ~ x1 + x2)
```

30. What does linearHypothesis() from the car package do with the model in this call?

```
library(car)
linearHypothesis(model, c("x1 = 0", "x2 = 0"))
```

31. What does anova(restricted, unrestricted) do, and what test is it here?

```
restricted <- lm(wage ~ educ + exper, data = df)
unrestricted <- lm(wage ~ (educ + exper)*female, data = df)
anova(restricted, unrestricted)
```

Econometrics Midterm — Model answers

1. W1 · Finding the mode with table() and sort()
table(): counts how many times each distinct value occurs and returns a **frequency** table (here 2 twice, 3 three times, 5 once)
sort(): with decreasing = TRUE orders those counts from largest to smallest (**descending**)
The first entry is the value with the highest count, the **mode** (here 3)
Key results: frequency · decreasing · mode

2. W2 · read.csv() and its arguments
read.csv(): reads a comma-separated text file and returns a **data frame**
header: TRUE means the **first row** holds the column (variable) names, not data
na.strings: every cell that contains "." is read as **NA** (missing), so it does not enter calculations as a value
Several codes can be given at once, for example c(".", "-")
Key results: data frame · first row · NA

3. W2 · attach() and detach()
attach(): adds the data frame to R's **search path**, so its columns can be used **by name** (COMPETENCE instead of practice$COMPETENCE)
detach(): **removes** the data frame from the search path again
Risk: a column can mask, or be masked by, another object with the same name
Key results: search path · by name · removes

4. W2 · quantile() and IQR()
quantile(): by default returns the 0%, 25%, 50%, 75% and 100% points, that is the **minimum**, the first **quartile**, the median, the third quartile and the maximum (0, 1, 2, 4.5, 12)
IQR(): the interquartile range, the **third quartile minus** the first quartile (4.5 - 1 = 3.5), the spread of the middle 50% of the data
Key results: quartile · minimum · third quartile minus

5. W2 · boxplot() and abline(h =)
boxplot(): draws a **box** from the lower to the upper hinge (about Q1 to Q3) with a bold line at the **median**
The **whiskers** reach the most extreme points within 1.5 times the IQR of the box, and points beyond them are drawn as outliers
abline(): with h set to a value it draws a **horizontal** line at that y-value, so here a red line through the median
Key results: box · median · whiskers · horizontal

6. W2 · vioplot() and its package
vioplot(): draws a violin plot for each variable, a smoothed **density** mirrored on both sides with a small **box plot** inside
install.packages(): downloads the **package** once, because vioplot is not part of base R
library(): **loads** the package in every new session so that vioplot() can be found
Key results: density · box plot · package · load

7. W2 · par(mfrow =)
par(): sets **graphics parameters**
mfrow: c(1, 2) splits the plotting window into **1 row and 2 columns**, and the following plots fill the panels row by row
Here boxplot(grade) and vioplot(grade) appear **side by side**
The layout stays in effect **until** it is reset, for example with par(mfrow = c(1, 1))
Key results: graphics parameter · 1 row and 2 columns · side by side · until

8. W2 · source() with echo = TRUE
source(): **runs** every command saved in the script file, from top to bottom
echo: TRUE also **prints** each command in the console before its output, as if it had been typed there
Key results: runs · prints

9. W2 · Sample variance written out
(competence - mean(competence))^2: squares every deviation from the mean, and sum() adds them up (about 124.07)
length(): returns n = 27, so the sum is divided by **n - 1** = 26
The result, about 4.77, is the **sample variance**, the same value that var(competence) returns
Key results: sample variance · n - 1 · var

10. W2 · Logical indexing with [ ]
am == 0 compares every element with 0 and returns a **logical** vector of TRUE and FALSE
mpg[...]: keeps **only** the mpg values at the positions where the condition is TRUE
mean(): averages those values, so the line gives the **average** mpg of **automatic** cars
attach(mtcars) is what lets mpg and am be used by name
Key results: logical · only · automatic · average

11. W2 · A user-defined function: to12()
to12(): converts an hour on the **24-hour** clock, given as text, into a 12-hour label
as.integer(): turns the text "22" into the **number** 22 so it can be compared with 12 and reduced
paste(): **joins** its arguments with a space, so to12("22") returns "10 PM"
Flaw: hour > 12 sends 12 to the else branch, so noon is returned as "12 AM"
Key results: 24-hour · number · joins · 10 PM

12. W2 · Why as.integer() is needed
hour: "13" is a **character** string, and arithmetic is not defined for text
So hour - 12 stops with an **error** (non-numeric argument to binary operator), even though the comparison as.integer(hour) > 12 works
to12() first **converts** the input with as.integer(), so hour - 12 is computed on the number 13
Key results: character · error · as.integer

13. W2 · str_detect() with fixed()
str_detect(): returns **TRUE** or FALSE for each string, depending on whether it **contains** the pattern
fixed(): makes the pattern a **literal** string, matched character by character, not a regular expression
So hasMultipleLoc("Seoul, Suwon") is TRUE and hasMultipleLoc("Seoul") is FALSE
Key results: TRUE · contains · literal

14. W2 · sapply()
sapply(): calls to12() on **each element** of data$openingHour and simplifies the results into a **vector**
It is needed because to12() uses **if**, which takes a single TRUE or FALSE, so calling to12() on the whole column at once fails
Key results: each element · vector · if

15. W2 · ifelse()
ifelse(): checks a **condition** for **each element** and returns the second argument where it is TRUE and the third where it is FALSE
Here the TRUE or FALSE from hasMultipleLoc() becomes "Yes" or "No", one per café, and is stored in the new column hasMultipleLoc
Key results: condition · each element · Yes

16. W2 · read_excel(), as.data.frame() and colnames()
read_excel(): from the readxl package, reads the **Excel** file Café.xlsx into a **tibble**
as.data.frame(): converts the tibble into an ordinary base R data frame
colnames(data) = c(...): **renames** the four columns to type, location, openingHour and closingHour
Key results: Excel · tibble · rename

17. W5 · lm() and abline() with a model
lm(): fits a **linear model** by OLS, perform on eye, and returns a model object that holds the coefficients
abline(): given that object, draws the **fitted regression line** (its intercept and slope) on the scatterplot, in blue
The model is **stored** in eye_regr so that later calls such as summary() and confint() can use it
Key results: linear model · regression line · stores

18. W5 · summary() of an lm model
summary(): for an lm object reports the **coefficient table**, with Estimate, Std. Error, t value and p-value for each term
It also gives the residual standard error, the **R-squared** and adjusted R-squared, and the overall **F-statistic** with its p-value
The $r.squared part extracts only the R-squared from that summary
Key results: coefficient table · t value · R-squared · F-statistic

19. W5 · confint() and its level argument
confint(): returns a **confidence interval** for each **coefficient** of the model, one row per term with a lower and an upper bound
By **default** the level is 95% (columns 2.5 % and 97.5 %)
level: 0.99 gives 99% intervals (columns 0.5 % and 99.5 %), which are **wider**
Key results: confidence interval · coefficient · 95% · wider

20. W5 · cor() and round()
cor(): computes the Pearson **correlation** between perform and skill, and ^2 squares it
In a simple regression the squared correlation equals the **R-squared**, which is what the script checks
round(): rounds the difference to 16 decimal places so that tiny **floating**-point differences between the two calculations are not mistaken for a real gap
Key results: correlation · R-squared · floating

21. W5 · predict() with interval = "confidence"
predict(): plugs the **new data**, a data frame with the same variable names as the model, into the fitted equation and returns the **predicted** values (about 0.3475 and 0.3441)
interval: "confidence" adds the **lower and upper** bounds, so the result is a matrix with the columns fit, lwr and upr
level: 0.95 sets the confidence level of those bounds
Key results: predicted · new data · lwr · 0.3475

22. W5 · seq(), polygon() and lines()
seq(): creates 10,000 **evenly spaced** x-values from min(eye) - 3 to max(eye) + 3
polygon(): joins the upper bounds (reversed) and the lower bounds into one closed shape and **fills** it in transparent grey, the **shaded** confidence band
lines(): draws the upper and lower bounds as green **dashed** lines
Key results: evenly spaced · fills · dashed

23. W5 · set.seed() and rnorm()
set.seed(): fixes the starting point of the random number generator, so the same random numbers, and the same results, are **reproducible** on every run
rnorm(): draws sampleSize values from a **normal** distribution, here with **mean** 10 and **standard deviation** 2 for fertilizer
The second rnorm(sampleSize, 0, 1) adds random errors with mean 0 and standard deviation 1
Key results: reproducible · normal · mean 10 · standard deviation

24. W5 · The for loop and the coefficient matrix
for: **repeats** the body 10,000 times, once per rowIdx, each time drawing a new sample and fitting lm()
summary(...)$coefficients: is the coefficient matrix, and row 2 is fertilizer
coef[2,1]: the slope **estimate**, coef[2,3] its **t value** and coef[2,4] its **p-value**, each stored at position rowIdx of its vector
Key results: repeats · estimate · t value · p-value

25. W5 · ifelse(), table() and barplot()
pValue <= alpha gives TRUE where the p-value is at most 0.05
ifelse(): turns TRUE and FALSE into the **labels** "Reject" and "Fail to Reject"
table(): **counts** how many of the 10,000 samples fall under each label
barplot(): draws one **bar** per label with its count
Because the true slope is 0.2, not 0, the share of Reject estimates the test's **power**
Key results: labels · counts · bar · power

26. W5 · Filling a results data frame
data.frame(): with index = 1:sampleCnt creates a **data frame** with 20 rows
results[rowIdx, "slope"] = ... writes into one **cell**, chosen by row number and column name, and creates the column the first time it is used
interval["fertilizer", "2.5 %"] picks the **lower** bound of the fertilizer interval from the confint() matrix, and "97.5 %" the upper bound
&& is TRUE only if **both** conditions hold, so inside is TRUE when the true slope lies within the interval
Key results: data frame · cell · lower · both

27. W5 · ggplot() layers
ggplot(): sets the data (results) and the mapping of slope to the x-axis and index to the y-axis
geom_point(): draws each estimated slope as a **point**, one row per sample
geom_errorbarh(): draws a horizontal **error bar** from lower_bound to upper_bound, the confidence interval
geom_vline(): adds a green **vertical line** at the true slope 2, so the intervals that miss it are easy to see
Key results: point · error bar · vertical line

28. W6 · as.factor() inside lm()
as.factor(): turns session into a **factor**, a categorical variable, instead of a number
lm(): then creates a **dummy** variable for every category except one, the **base** (reference) category, which is absorbed in the intercept
Each dummy coefficient is the difference from the base category, holding x constant
Key results: factor · dummy · base

29. W6 · The : and * operators in a formula
x1:x2 adds only the **interaction** term, the product of x1 and x2
x1*x2: is shorthand for x1 + x2 + x1:x2, so it adds **both** the **main effects** and the interaction
x1 + x2: has the main effects only
Key results: interaction · main effects · both

30. W6 · linearHypothesis()
linearHypothesis(): from the car package, tests linear **restrictions** on the coefficients of a fitted model
Here it tests "x1 = 0" and "x2 = 0" **jointly**, not one at a time
For an lm model it reports an **F-test**, with the residual sums of squares of the restricted and unrestricted models, F and its p-value
Key results: restrictions · jointly · F-test

31. W6 · anova() with two nested models
anova(): given two **nested** models, **compares** them with an **F-test**
It uses the residual sum of squares (**RSS**) of each model and the number of extra parameters, and reports F and its p-value
Here the unrestricted model interacts every term with female, so the test is the **Chow** test of whether one regression fits both groups
Key results: compares · F-test · RSS · Chow